SAFETY DATA SHEET

LAUNDRY SOUR 500

1. Chemical product and company identification

Product name: LAUNDRY SOUR 500
Code: 972257
Recommended use and restrictions: Laundry product.
Use only for the purpose on the product label.

Supplier address
Ecolab Food Safety & Hygiene Solutions Pvt. Ltd.
Unit No. 808, "215-Atrium",8th floor,
Next to Courtyard by Marriott Hotel, Andheri Kurla Road,
Andheri (East), Mumbai - 400 059, India
Toll free number: 1800 209 2530

Emergency telephone number: 1-651-222-5352 (in USA)

2. Hazards identification

Product hazard classification:
ACUTE TOXICITY: ORAL - Category 5
SKIN CORROSION/IRRITATION - Category 1A
SERIOUS EYE DAMAGE/ EYE IRRITATION - Category 1

<u>Label contents</u>

Symbol:

Signal word: DANGER
Hazard warning message: May be harmful if swallowed.
Causes severe skin burns and eye damage.

Precautionary statements
Prevention: Wear protective gloves. Wear eye/face protection. Wash thoroughly after handling.
Response: Immediately call a POISON CENTER or doctor/physician. IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower. Wash contaminated clothing before reuse. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. IF INHALED: Remove to fresh air and keep at rest in a position comfortable for breathing.
Other hazards: Not known.

3. Composition/information on ingredients

Nature of material: Mixture.
Chemical properties: Laundry product.

Hazardous ingredients	Concentration Range (%)	CAS number
formic acid	20 – 50	64-18-6

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment and hence require reporting in this section.

4. First-aid measures

First aid measures for different exposure routes

Eye contact: In case of contact, immediately flush eyes with cool running water. Remove contact lenses and continue flushing with plenty of water for at least 15 minutes. Get medical attention immediately.

Skin contact: In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Get medical attention immediately. Wash clothing before reuse. Clean shoes thoroughly before reuse.

Inhalation: If inhaled, remove to fresh air. If exposed person is not breathing, give artificial respiration or oxygen applied by trained personnel. Get medical attention immediately.

Ingestion: If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Do not induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

Most important symptoms/effects, acute and delayed

Eye contact: Causes serious eye damage.

Skin contact: Causes severe burns.

Inhalation: May give off gas, vapor or dust that is very irritating or corrosive to the respiratory system.

Ingestion: May be harmful if swallowed. May cause burns to mouth, throat and stomach.

Protection of first-aiders: No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

Notes to physician: No specific treatment. Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.

See toxicological information (section 11).

5. Fire-fighting measures

Suitable fire extinguishing media: Use water spray, fog or foam.

Specific hazards that may be encountered during fire- In a fire or if heated, a pressure increase will occur and the container may burst.

Hazardous thermal decomposition products: Decomposition products may include the following materials:
carbon dioxide
carbon monoxide

Specific fire-fighting methods: Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

Special protective equipment for fire-fighters: Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

6. Accidental release measures

Personal precautions: Immediately contact emergency personnel. Stop leak if without risk. Use suitable protective equipment. Keep unnecessary personnel away. Do not touch or walk through spilled material.

Environmental precautions: Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

Methods for cleaning up: If emergency personnel are unavailable, contain spilled material. For small spills, add absorbent (soil may be used in the absence of other suitable materials), scoop up material and place in a sealable, liquid-proof container for disposal. For large spills, dike spilled material or otherwise contain it to ensure runoff does not reach a waterway. Place spilled material in an appropriate container for disposal. Contaminated absorbent material may pose the same hazard as the spilled product.

7. Handling and storage

Handling: Do not ingest. Do not get in eyes or on skin or clothing. Avoid breathing vapors or mists. Use only with adequate ventilation. Wash thoroughly after handling.

Storage: Keep out of reach of children. Keep container tightly closed. Keep container in a cool, well-ventilated area.
Do not store above the following temperature: 50°C

8. Exposure controls/personal protection

Control parameters / Occupational exposure limits:

Ingredient name	Exposure limits
Formic acid	**Singapore WHS (General Provisions) Regulations 2006, First Schedule.** PEL long term 9.4 mg/m³. PEL short term 19 mg/m³.

Biological indicator: Not available.

Appropriate engineering controls: Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Individual protection measures

Hygiene measures: Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing.

Eye protection: Use chemical splash goggles. For continued or severe exposure wear a face shield over the goggles.

Hand protection: Use chemical-resistant, impervious gloves.

Skin protection: Use synthetic apron, other protective equipment as necessary to prevent skin contact.

Respiratory protection: A respirator is not needed under normal and intended conditions of product use.

9. Physical and chemical properties

Appearance

Physical state: Liquid.
Color: Colorless.
Odor: Pungent.
Odor threshold: Not available.
pH: 1 to 2 [Conc. (% w/w): 100%].
Melting point: Not available.
Initial boiling point: > 100°C.
Flash point: Not available.
Evaporation rate (butyl acetate = 1): Not available.
Flammability (solid, gas): Not available.
Explosion limits: Not available.
Vapor pressure: Not available.
Vapor density: Not available.
Relative density: 1.05 to 1.08 (Water = 1) @20°C.
Solubility: Soluble in water in all proportions.

Octanol/water partition coefficient: Not available.
Auto-ignition temperature Not available.

10. Stability and reactivity

Stability: The product is stable.
Possibility of hazardous reactions: Under normal conditions of storage and use, hazardous reactions will not occur.

Conditions to avoid: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Materials to avoid: Highly reactive or incompatible with the following materials: alkalis.
Reactive or incompatible with the following materials: metals.
Slightly reactive or incompatible with the following materials: moisture.
Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Hazardous decomposition products: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

11. Toxicological information

Route of exposure: Skin contact, Eye contact, Inhalation, Ingestion.

<u>**Symptoms**</u>

Eye contact: Adverse symptoms may include the following:
pain
watering
redness

Skin contact: Adverse symptoms may include the following:
pain or irritation
redness
blistering may occur

Inhalation: Adverse symptoms may include the following:
.
coughing
Respiratory tract irritation

Ingestion: Adverse symptoms may include the following:
stomach pains

<u>**Acute toxicity**</u>

Eye contact: Causes serious eye damage.
Skin contact: Causes severe burns.
Inhalation: May give off gas, vapor or dust that is very irritating or corrosive to the respiratory
Ingestion: May be harmful if swallowed. May cause burns to mouth, throat and stomach.

<u>**Toxicity data**</u>

Product/ingredient name	Result	Species	Dose	Exposure
formic acid	LD_{50} Oral	Mouse	1076 mg/kg	-
	LD_{50} Oral	Mouse	700 mg/kg	-
	LD_{50} Oral	Rat	1830 mg/kg	-
	LD_{50} Oral	Rat	1100 mg/kg	-
	LC_{50} Inhalation Vapor	Rat	15 g/m^3	15 minutes

<u>**Chronic toxicity**</u>

Carcinogenicity: No known significant effects or critical hazards.
Mutagenicity: No known significant effects or critical hazards.
Teratogenicity: No known significant effects or critical hazards.
Developmental effects: No known significant effects or critical hazards.
Fertility effects: No known significant effects or critical hazards.

12. Ecological information

Ecotoxicity: This material is toxic to aquatic life.

Aquatic and terrestrial toxicity

Product/ingredient name	Test	Result	Species	Exposure
formic acid	Intoxication	Acute EC50 151.2 mg/L	Daphnia	48 hours

Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
Not available.				

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
Not available.			

Bioaccumulative potential

Product/ingredient name	$LogP_{ow}$	BCF	Potential
Not available.			

Soil mobility: Not available.
Other adverse effects: No known significant effects or critical hazards.

13. Disposal considerations

Disposal methods: The generation of waste should be avoided or minimized wherever possible. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe way. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

14. Transport information

UN number:	UN3412
Proper shipping name:	FORMIC ACID SOLUTION
Class:	8
Packing group:	II
Marine pollutant (Yes / No):	Not a pollutant.
Specific precautionary transport measures and conditions:	-

15. Regulatory information

Applicable Singapore regulations:

Workplace Safety and Health Act & Workplace Safety and Health (General Provisions) Regulations: This product is subject to the SDS, labeling, PEL and other requirements in the Act/Regulations.

Environmental Protection and Management Act and Environmental Protection and Management (Hazardous Substances) Regulations: This product is subject to the requirements in the Act/Regulations.

Maritime and Port of Singapore (Dangerous Goods, Petroleum and Explosives) Regulations: This product is subject to the requirements of the Regulations.

This product is not classified as Flammable Material under the Safety Act and Fire Safety (Petroleum and Flammable Materials) Regulations 2005.

This product is classified as Hazardous Substance according to the Environmental Protection and Management (Hazardous Substances) Regulations.

This product is not classified as Explosive Precursors under the Arms & Explosives Act.

16. Other information

History

Date of issue:	July 20, 2009
Date of previous issue:	**No previous validation.**
Organization that prepared the MSDS:	Ecolab Pte Ltd
Person who prepared the MSDS:	Technical Center

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.